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N/a Nikhil
s224234862
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Tutor:
Ananda Maiti

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SIT746 Research Project (Advanced)

Task 2.1D — Complete Literature Review

GPS Spoofing Detection in Autonomous Mobile Robots: A Critical Literature Review

Student Name:	Nikhil Sarwara
Student ID:	224234862
Supervisor:	Dr. Ananda Maiti
Unit:	SIT746 Research Project (Advanced)
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Deakin University, School of Information Technology

Abstract

Autonomous mobile robots rely heavily on Global Navigation Satellite Systems (GNSS), particularly GPS, for localisation, trajectory tracking, and autonomous mission execution. This dependency creates a critical cybersecurity vulnerability: GPS spoofing, in which counterfeit satellite signals manipulate the receiver's estimated position, velocity, or time. This literature review critically examines research on GPS spoofing in autonomous mobile robots, with particular attention to resource-constrained robotic platforms such as unmanned aerial vehicles (UAVs) and other embedded navigation systems. The review is structured around three research questions: (1) the primary vulnerabilities of GPS in autonomous mobile robots; (2) the effectiveness of real-time sensor fusion techniques for spoofing detection; and (3) the design need for a dynamic GPS Trust Index that moves beyond binary spoofed/authentic outputs. The review synthesises literature on signal vulnerabilities, attack taxonomies, signal-level and receiver-level detection, GPS–IMU and multi-sensor fusion, machine learning approaches, and simulation-based evaluation frameworks. It identifies three persistent gaps: limited platform-specific vulnerability analysis for mobile robots, insufficient real-time validation of sensor fusion under practical robotic constraints, and the absence of standardised continuous trust quantification for GPS reliability. These gaps provide the basis for the present research direction, which focuses on a real-time, simulation-driven, sensor-fusion-based spoofing detection framework for autonomous mobile robots.

Keywords: GPS spoofing, GNSS security, autonomous mobile robots, sensor fusion, IMU, real-time detection, trust index, UAV cybersecurity, simulation-based evaluation.

Contents

Introduction

Background and Motivation

Autonomous mobile robots, including UAVs, ground robots, and marine platforms, depend on accurate positioning to perform navigation, localisation, waypoint tracking, collision avoidance, and mission planning. In many practical systems, GPS remains the dominant source of global position information because it is widely available, inexpensive, and already integrated into most commercial autopilot and robotic navigation stacks [1, 2].

However, civilian GPS is fundamentally vulnerable. The received signal is weak, the civil signal structure is publicly documented, and most civilian receivers lack message authentication. These properties make GPS susceptible to spoofing, where an attacker transmits counterfeit signals that appear legitimate to the receiver and gradually or abruptly manipulate the estimated position solution [1, 6]. For autonomous robots, the consequences are more serious than for static receivers because a compromised navigation solution can propagate directly into control decisions, trajectory updates, and safety functions [2, 14].

GPS spoofing has therefore become both a cybersecurity problem and a cyber-physical systems problem. It cannot be treated only as a radio-frequency anomaly, because its impact emerges through the interaction between sensing, estimation, control, and mission execution. This is particularly important in embedded robotic systems that must detect anomalies under tight latency, compute, and power constraints [2, 3].

Scope of This Review

This review focuses on *detection* rather than cryptographic prevention. Specifically, it reviews literature relevant to detecting GPS spoofing in autonomous mobile robots using signal-level analysis, sensor fusion, machine learning, and simulation-based evaluation. The review includes UAVs, ground robots, and related mobile autonomous platforms because they share common resource constraints and motion-dependent sensing challenges. At the same time, the evaluative focus is deliberately placed on robotic and embedded deployment contexts rather than aviation-grade or laboratory-only GNSS receivers.

The review primarily examines literature published between 2019 and 2026, while including a small number of foundational earlier works where necessary for core concepts such as GPS spoofing theory and GNSS authentication. This time window was selected to capture current research developments while maintaining direct relevance to contemporary autonomous robotic platforms.

Review Methodology

This literature review was conducted as a structured thematic review rather than a formal systematic review. Relevant sources were identified through searches of major academic databases and scholarly search engines, including IEEE Xplore, Scopus, ScienceDirect, SpringerLink, and Google Scholar. Search terms included combinations of *GPS spoofing*, *GNSS spoofing detection*, *autonomous robots*, *UAV spoofing*, *sensor fusion*, *IMU*, *trust model*, *machine learning*, and *simulation framework*.

Priority was given to peer-reviewed journal articles, conference papers, recognised dissertations, and technically substantial framework papers. Sources were included when they addressed at least one of the following: GPS signal vulnerabilities, spoofing detection techniques, robotic or UAV deployment constraints, trust or reliability assessment, or simulation and evaluation frameworks. Purely cryptographic signal redesign studies, military-only signal protection work without relevance to civilian robotic systems, and papers lacking clear methodological detail were treated as peripheral rather than central to the analysis.

Research Questions

The review is organised around the following research questions:

RQ1: What are the primary vulnerabilities of GPS in autonomous mobile robots to spoofing attacks?

RQ2: How can sensor fusion techniques effectively detect GPS spoofing in real time?

RQ3: How can a dynamic GPS Trust Index be designed to quantify the reliability of GPS data?

Theme Alignment

Table 1: Alignment Between Research Questions and Literature Themes

RQ	Primary Theme	Focus of Review
RQ1	GPS vulnerabilities and attack taxonomies	Civilian GPS weaknesses, spoofing modes, platform-specific vulnerabilities in autonomous mobile robots
RQ2	Detection techniques and simulation frameworks	Signal-level detection, GPS–IMU fusion, multi-sensor fusion, machine learning, real-time deployment constraints, simulation-based validation
RQ3	Trust and reliability assessment	Binary versus continuous detection, trust modelling, dynamic reliability scoring, navigation-aware GPS confidence assessment

Structure of the Review

The remainder of the review is organised as follows. Section 2 examines GPS vulnerabilities and attack taxonomies in relation to RQ1. Section 3 reviews detection techniques, with emphasis on real-time sensor fusion and machine learning relevant to RQ2. Section 4 discusses simulation and evaluation frameworks that support practical real-time validation. Section 5 analyses the literature on trust and reliability assessment in relation to RQ3. Section 6 provides an integrative synthesis and maps the major research gaps directly to the research questions. Section 7 concludes the review.

GPS Vulnerabilities and Attack Taxonomies

Fundamental Weaknesses of Civilian GPS

Civilian GPS signals possess several structural weaknesses that make spoofing possible. The signals arrive at extremely low power, operate in open broadcast form, and lack strong authentication in most civilian deployments. As a result, a counterfeit signal with sufficient power and protocol conformity can capture the receiver's tracking loops and distort the position solution without requiring physical access to the platform [1, 6].

This vulnerability is not equally severe across all systems. In stationary or high-grade receivers, additional filtering, signal-quality monitoring, or multi-frequency hardware may reduce exposure. In contrast, autonomous mobile robots often rely on cost-sensitive, commercial-off-the-shelf GNSS modules with limited access to lower-level signal diagnostics, creating a narrower defensive surface [2, 14]. The literature therefore supports the view that GPS vulnerability in robotics is not merely a general GNSS issue, but a deployment-specific issue shaped by cost, sensing architecture, and embedded constraints.

Spoofing Attack Taxonomies

The spoofing literature commonly distinguishes between abrupt attacks and gradual attacks. Sudden-jump attacks create immediate discontinuities in reported position or velocity and are comparatively easier to detect. Gradual-drift attacks are more subtle: the counterfeit solution is moved incrementally so that the manipulated trajectory remains superficially plausible over short time windows [1, 6]. The latter is especially important for autonomous robots because small navigation deviations can accumulate into major mission errors before triggering obvious alarm conditions.

More advanced taxonomies also distinguish between meaconing or replay-style attacks, coordinated multi-satellite spoofing, and composite attacks that combine spoofing with jamming or other cyber-physical disturbances [7, 15]. These categories matter because detection assumptions change across attack types. A method effective against abrupt jumps may fail against smooth drift, while a GPS–IMU consistency check may be weakened if multiple subsystems are jointly disrupted.

Platform-Specific Vulnerabilities in Autonomous Mobile Robots

The literature shows that autonomous mobile robots face a distinct vulnerability profile compared with general GNSS receivers. First, mobile robots are dynamic systems, so their sensor readings naturally vary due to manoeuvres, vibration, terrain interaction, and environmental disturbances. This makes it harder to separate malicious anomalies from normal motion [2, 3]. Second, embedded robotic platforms must operate within strict compute, memory, and latency budgets. A detection method that performs well in offline analysis or on desktop hardware may not be deployable in a real-time robotic stack [2, 8].

Third, robots are decision-making systems rather than passive receivers. A spoofed position estimate can alter control commands, route planning, obstacle avoidance behaviour, or failsafe logic. In this sense, GPS spoofing in robotics is not just a localisation error; it is a control-layer threat. Yaacoub et al. show that such robotic vulnerabilities should be understood as part of a broader cyber-physical attack surface rather than as an isolated RF-layer problem [14].

Critical Evaluation for RQ1

The literature clearly identifies the general weaknesses of civilian GPS, but it is less mature in analysing how those weaknesses manifest in specific mobile robotic contexts. Many papers discuss GNSS vulnerability at a broad level, yet provide limited treatment of resource-constrained robotic platforms that must make control decisions in real time. This means that the foundational theory of GPS weakness is well established, but the platform-specific characterisation required by RQ1 remains incomplete.

A further limitation is that vulnerability discussions often over-emphasise signal characteristics while under-emphasising system architecture. In practice, the severity of spoofing depends not only on the signal but also on whether the robot has IMU support, odometry, optical flow, barometric sensing, robust state estimation, or fallback navigation logic. The literature therefore supports the need for a more integrated vulnerability analysis tailored to autonomous mobile robots rather than general GNSS receivers alone.

Detection Techniques for Real-Time GPS Spoofing Detection

Signal-Level and Receiver-Level Detection

Signal-level detection methods attempt to identify spoofing by analysing received power, signal quality, correlation distortion, or automatic gain control behaviour. These methods are appealing because they target the spoofing process close to its source and can be effective against unsophisticated attacks [1, 7]. However, their practical usefulness in autonomous mobile robots is constrained by a major implementation problem: most commercial robotic platforms do not expose the correlator outputs or low-level receiver internals needed for advanced signal-quality monitoring.

This creates a mismatch between much of the signal-level literature and the deployment reality of embedded robotic systems. Methods evaluated with bench-top signal generators or specialised receivers may not transfer to PX4-class, ArduPilot-class, or other companion-computer-based platforms where only higher-level telemetry is available [8, 13]. From a robotic systems perspective, signal-level detection is therefore important but often insufficient as a standalone practical solution.

GPS–IMU Sensor Fusion

Sensor fusion is one of the most promising directions for real-time spoofing detection because it uses internal consistency rather than absolute signal trust. The underlying idea is that GPS-reported motion should be consistent with motion inferred from independent sensors such as IMUs, odometry, barometers, or optical flow. If GPS indicates movement or trajectory change that is not supported by the robot's physical sensing, spoofing becomes a plausible explanation [3, 4].

Dasgupta et al. present one of the strongest examples of this approach, using an Extended Kalman Filter (EKF) and innovation residual analysis to compare GPS measurements against inertial predictions [3]. This approach is methodologically strong because it embeds spoofing detection within the normal state-estimation process rather than treating it as a completely separate task. It is also more compatible with autonomous systems because it leverages data that many platforms already produce.

At the same time, fusion-based detection has limitations. IMUs drift over time, visual sensors are sensitive to environmental texture and lighting, and tightly coupled fusion may require raw measurements unavailable on low-cost hardware [4, 8]. In addition, many reported evaluations rely on abrupt attack scenarios or limited datasets, leaving open the question of how robust these methods remain under gradual drift or changing mission conditions.

Other Multi-Sensor Consistency Approaches

Beyond GPS–IMU fusion, some studies use optical flow, barometric altitude, vehicle dynamics, or cross-vehicle information to strengthen detection. Optical flow can be useful in UAVs because it provides an additional motion estimate that does not rely on GNSS, but it introduces new payload, compute, and environmental dependencies [4]. Barometric consistency can support altitude-related checks at low cost, though it is generally a secondary rather than primary spoofing indicator [11]. Collaborative or federated methods may improve fleet-level robustness, but they assume communications infrastructure that may not exist in single-robot or disconnected settings [10, 15].

These extensions show that sensor fusion is best viewed as a family of approaches rather than a single method. Their effectiveness depends on which signals are available, how tightly they are integrated, and whether the method is designed for the operational realities of autonomous robots rather than idealised testing conditions.

Machine Learning and Deep Learning Approaches

Machine learning methods have gained prominence because spoofing often manifests through complex temporal and cross-feature patterns that are difficult to capture with fixed rules alone. Supervised models such as Support Vector Machines, Random Forests, and neural networks have all been applied to GPS spoofing detection [3, 9]. Their appeal lies in their ability to learn nonlinear relations among signal quality, motion behaviour, and estimator consistency features.

Shafique et al. show that machine learning can achieve high classification accuracy, but their evaluation also illustrates a common problem in the literature: model results are often reported without sufficient attention to platform realism, dataset diversity, or motion complexity [9]. Similarly, deep learning approaches such as CNNs and LSTMs may improve temporal pattern recognition, yet they increase computational burden and can be difficult to justify on resource-constrained embedded systems [3, 10].

A recurring weakness is that reported accuracy figures are often not directly comparable. Some studies use stationary or semi-simulated datasets, some use proprietary traces, and some do not report real-time latency in enough detail to judge embedded feasibility. As a result, the machine learning literature is promising but methodologically uneven when assessed against RQ2, which requires not only detection accuracy but *effective real-time operation*.

Critical Evaluation for RQ2

The literature supports sensor fusion as the most practically relevant direction for real-time robotic spoofing detection, especially when compared with purely signal-level approaches that require inaccessible receiver internals. However, there remains a clear gap between proof-of-concept studies and embedded, real-time robotic deployment. Many approaches are validated in simulation, offline replay, or non-robotic domains, while relatively few studies demonstrate how detection is integrated into a realistic navigation stack under real-time constraints [3, 13].

Another important limitation is that detection is often evaluated as a binary classification task detached from control and decision-making. For autonomous robots, the practical question is not only whether spoofing is detected, but whether it is detected early enough, reliably enough, and cheaply enough to support safe navigation. This is where RQ2 remains open and why sensor fusion needs to be examined in terms of both algorithmic strength and implementation feasibility.

Simulation and Evaluation Frameworks

Why Simulation Matters

Simulation plays a central role in GPS spoofing research because real-world RF spoofing experiments are difficult, costly, and potentially unsafe. In robotic security research, simulation makes it possible to generate controlled attack conditions, collect repeatable telemetry, and vary spoofing parameters without interfering with real navigation systems [12, 13].

For autonomous mobile robots, this is especially useful because detection methods must be tested across multiple trajectories, noise conditions, motion profiles, and attack intensities. These requirements are difficult to satisfy with real-world experiments alone. Simulation therefore supports both methodological reproducibility and broader scenario coverage.

Gazebo, ROS2, and UAV Security Frameworks

Recent work has begun to move from abstract spoofing models toward full robotic simulation frameworks. Pekaric et al. show that Gazebo can be used to simulate sensor spoofing scenarios while preserving control over ground-truth state variables [12]. Patil et al. extend this idea by presenting a ROS2-based cyber-physical security framework for UAV testing, which is valuable because it treats spoofing detection as part of a full system evaluation process rather than as an isolated classification task [13].

These frameworks are important for RQ2 because they enable practical testing of real-time detection logic using robot-relevant telemetry. They also reveal a key engineering truth: evaluation environments must reflect the interfaces that real robotic systems actually expose, such as MAVLink telemetry, sensor topics, and autopilot status outputs, rather than assuming access to idealised internal signals.

Limitations of Current Evaluation Practice

Despite progress, current evaluation practice still has several limitations. First, many studies rely on narrow attack scenarios, often favouring sudden position jumps over more difficult gradual drift. Second, datasets are frequently small, proprietary, or insufficiently described for reproduction [9, 10]. Third, simulation-based results do not automatically transfer to real-world platforms because of sim-to-real gaps involving sensor noise, vibration, multipath, and environmental disturbances [4].

A further weakness is that real-time claims are often under-specified. Some papers report high accuracy without carefully quantifying processing latency, windowing delay, computational cost, or integration overhead. For robotic navigation systems, these details are essential. A detector that is accurate but too slow may still be operationally inadequate. This is one of the major reasons why real-time validation remains an open issue under RQ2.

Trust and Reliability Assessment

Limitations of Binary Detection Outputs

A major pattern in the literature is the dominance of binary outputs: *spoofed* or *authentic*. While this is convenient for benchmarking, it is not always sufficient for autonomous robots that must decide how much to trust GPS at each moment and whether to reweight, ignore, or downgrade the GPS measurement within a broader sensor fusion pipeline [5, 11].

Binary classification is therefore a limited abstraction. In practical systems, spoofing may unfold progressively, uncertainty may increase before a hard alarm is justified, and navigation controllers may need gradual adaptation rather than abrupt switching. This motivates the shift from detection as classification to detection as reliability assessment.

Trust-Based Perspectives in the Literature

Trust-based models attempt to represent confidence in a sensor or system as a continuous quantity rather than a binary state. Azevedo-Sa et al. show how continuous trust modelling can support smoother system behaviour in automated driving contexts [5]. Tariq and Tariq similarly demonstrate that trust-informed anomaly reasoning can reduce false alarms by combining multiple signal cues into a graded reliability estimate [11].

Although these studies are not all situated directly within autonomous mobile robot spoofing detection, they are highly relevant to RQ3 because they establish the conceptual need for a dynamic reliability variable. They suggest that GPS trust should be updated over time based on anomaly evidence, signal quality, and historical consistency rather than assigned once as a fixed binary label.

Gap in Dynamic GPS Trust Quantification

The trust literature remains underdeveloped for robotic spoofing detection in three ways. First, there is no widely adopted standard for how GPS trust should be quantified in autonomous systems. Second, existing trust models often emerge from adjacent domains such as automated driving or VANET security, where system assumptions differ from single-robot navigation. Third, few studies explain how a continuous trust estimate should be integrated with real-time sensor fusion in embedded robotic systems [5, 11].

This means that RQ3 addresses a genuine gap. The literature provides conceptual support for trust-aware detection, but it does not yet provide a standard, robot-ready method for continuous GPS reliability assessment under spoofing conditions. A dynamic trust index is therefore not simply an enhancement to binary detection; it is a missing bridge between anomaly detection and navigation-aware decision-making.

Critical Evaluation for RQ3

Compared with the literature on spoofing detection itself, the literature on continuous trust is smaller and less standardised. Yet it may be one of the most important directions for practical robotic systems. A detector that merely says "spoofed" offers limited operational value if it cannot inform how the robot should adapt its navigation logic. Continuous trust, by contrast, supports graded response, sensor reweighting, and smoother transition to fallback modes.

The challenge is that trust models can easily become vague if they are not tied to measurable evidence. For RQ3, the key requirement is therefore not only to define a trust index, but to ground it in observable sensor inconsistencies and real-time computable signals. The literature strongly motivates this direction, but does not yet resolve it.

Synthesis and Research Gaps

Comparative Synthesis

Table 2: Comparative Synthesis of Representative Literature

Study	Primary Focus	Platform	Real-Time Relevance	Main Limitation for This Review
Ahmad et al. [1]	GPS weakness and detection overview	General GNSS	Indirect	Strong on fundamentals, but not specific to autonomous mobile robots
Psiaki and Humphreys [6]	GNSS spoofing theory and detection concepts	General GNSS	Indirect	Foundational but not focused on embedded robotic deployment
Kadena et al. [2]	Security challenges in mobile robots	Mobile robots	Moderate	Broad systems view, but limited implementation detail on spoofing detection
Dasgupta et al. [3]	GPS–IMU fusion and ML detection	Autonomous vehicles	High	Strong real-time relevance, but not centred on resource-constrained robotic autopilots
Wang et al. [8]	Multi-sensor fusion survey	Automated driving	Moderate	Valuable taxonomy, but many methods assume richer hardware access

Study	Primary Focus	Platform	Real-Time Relevance	Main Limitation for This Review
Meng et al. [4]	Optical-flow fusion for UAV spoofing detection	UAV	High	Adds useful modality, but increases payload and environmental dependence
Shafique et al. [9]	ML-based spoofing classification	UAV-related	Moderate	Reported performance is sensitive to dataset realism and mobility assumptions
Dang [10]	Transformer-based GNSS anomaly detection	UAV / cellular-connected systems	Moderate	Advanced ML framing, but proprietary data and embedded feasibility remain unclear
Tariq and Tariq [11]	Trust-aware anomaly detection	Vehicular network context	Moderate	Useful trust perspective, but not directly tailored to single-robot fusion pipelines
Pekaric et al. [12]	Gazebo-based spoofing simulation	UAV	Indirect to High	Valuable for reproducibility, but simulation results alone do not close the sim-to-real gap
Patil et al. [13]	ROS2-based UAV security framework	UAV	High	Strong framework orientation, but not itself a complete detection solution
Yaacoub et al. [14]	Cyber-physical robotics security survey	Robotics	Indirect	Good systems context, but limited detection-specific detail

The synthesis reveals that the literature is relatively mature in identifying individual components of the problem, but less mature in integrating them into a single robotic detection framework. GPS vulnerability is well understood at a signal level, sensor fusion is widely recognised as promising, and trust-based reasoning is conceptually established. However, the combination of these three elements in a practical, real-time, robot-oriented framework remains incomplete.

Gap 1: Limited Platform-Specific Vulnerability Characterisation (RQ1)

The first research gap is that GPS vulnerability is often discussed in generic GNSS terms rather than in platform-specific robotic terms. Autonomous mobile robots differ from aviation-grade or lab-grade systems because they are compute-limited, motion-dependent, and directly control physical behaviour. Existing literature does not yet provide a sufficiently detailed account of how spoofing vulnerabilities manifest under these robotic constraints. This gap maps directly to RQ1.

Gap 2: Insufficient Real-Time Validation of Sensor Fusion (RQ2)

The second gap concerns the move from promising methods to practical deployment. Many sensor-fusion and ML approaches achieve strong reported performance, but their real-time operation in realistic robotic navigation settings is under-validated. Common limitations include dependence on inaccessible low-level receiver data, narrow attack scenarios, insufficient latency reporting, and weak alignment with actual embedded system interfaces. This gap maps directly to RQ2.

Gap 3: Absence of Standardised Continuous GPS Trust Quantification (RQ3)

The third gap is the absence of a standardised, robot-oriented framework for continuous GPS trust assessment. The literature contains useful trust concepts, but no common method for converting anomaly evidence into a dynamic GPS reliability score that can inform real-time robotic navigation decisions. This gap maps directly to RQ3.

Implications for the Present Research Direction

Taken together, these gaps justify a research direction that integrates three elements: platform-aware vulnerability analysis, real-time sensor-fusion-based spoofing detection, and a dynamic trust-oriented interpretation of GPS reliability. Importantly, the literature suggests that none of these elements is sufficient alone. Vulnerability analysis without detection remains descriptive; detection without trust remains operationally incomplete; and trust without grounded sensor evidence risks becoming too abstract for engineering use.

Conclusion

This literature review has examined GPS spoofing research through the lens of three research questions concerned with vulnerability, real-time detection, and continuous trust assessment in autonomous mobile robots. The review shows that civilian GPS remains structurally vulnerable, that autonomous robots intensify the consequences of spoofing through tight coupling between sensing and control, and that current literature still under-characterises these vulnerabilities in robot-specific terms.

The review also finds that sensor fusion is the most practically relevant direction for real-time spoofing detection, especially where low-level receiver signals are inaccessible. However, existing literature does not yet provide enough evidence that current fusion and machine learning approaches are robust, low-latency, and fully validated for realistic embedded robotic deployment. Simulation frameworks have improved the situation, but they do not eliminate the need for careful real-time and system-level evaluation.

Finally, the review identifies a clear conceptual and methodological gap in the area of continuous GPS trust assessment. Most studies still treat spoofing detection as a binary classification problem, whereas autonomous mobile robots require graded and time-varying confidence in GPS data to support safe navigation. This makes the design of a dynamic GPS Trust Index a well-justified research direction rather than a peripheral extension.

Overall, the literature supports a strong integrated research agenda: understand robot-specific GPS vulnerabilities, evaluate real-time sensor fusion under practical constraints, and translate anomaly evidence into continuous reliability assessment. These three strands provide the intellectual foundation for the subsequent methodology and implementation stages of the project.

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